

Statistical Physics

Chengyu Jin

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Capítulo 1

Classical atomic gas

1.1. Ideal systems

In an ideal system I can express the system's hamiltonian as

$$H = \sum_i h_i \quad (1.1)$$

Applying this in the canonical ensemble framework's probability we have

$$p(C) = \frac{e^{\beta H(C)}}{Z} = \frac{e^{\beta \sum_i h_i}}{Z} = \prod_i \frac{e^{\beta h_i}}{Z} \quad (1.2)$$

Nota

The previous result is consistence with the stadistical result: if $A \cap B = 0$ then $P(A \cup B) = P(A)P(B)$

1.1.1. Examples

If we have a potential $V(r_{ij})$, this is a potential that depends on the relative position of the particle i with particles j. I cannot separate the hamiltonian because the particle i dependes on surrounding particles.

$$V(r_{ij}) \implies H \neq \sum_i h_i \quad (1.3)$$

If we have a potential $V(r_i)$, this is a potential that only depends on the particle's position. Then I can separate the hamiltonian.

$$V(r_i) \implies H = \sum_i h_i = \sum_i \frac{p_i^2}{2m_i} + \sum_i V(r_i) \quad (1.4)$$

The Ising Model that we have seen in Computational Physics has the following hamiltonian

$$H = -\frac{J}{2} \sum_{ij}^N s_{ij} (s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1}) \quad (1.5)$$

As we can see in the equation, the hamiltonian of the particle i-j depend on the surrounding, then we cannot express the hamiltonian as sums of each particles' hamiltonians.

$$H = -\frac{J}{2} \sum_{ij}^N s_{ij} (s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1}) \neq \sum_i h_i \quad (1.6)$$

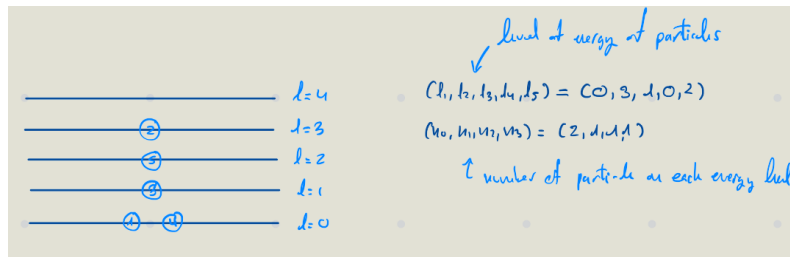
If we have the following hamiltonian, we can separate it in sum of particles' hamiltonian with $h_i = -Js_i$.

$$H = -J \sum_i s_i = \sum_i h_i = \sum_i -Js_i \quad (1.7)$$

1.2. Boltzmann Gas

The Boltzmann Gas it is the common limit of fermions and bosons.

1.2.1. In canonical ensemble



$$Z = \sum_{\{n_i\}, \sum_i n_i = N} e^{-\beta E} = \sum_{\{n_i\}, \sum_i n_i = N} e^{-\beta \sum_i n_i \epsilon_i} \quad (1.8)$$

Due to the condition $\sum_i n_i = N$, we cannot calculate the partition function as the sum of partition functions. But we can express the total energy as sum of energy associated to level l .

$$E = \sum_i n_i \epsilon_i = \sum_i E_{l_i} \quad (1.9)$$

Applying this in the partition function we obtain

$$Z = \sum_{l_1, \dots, l_N} e^{\beta \sum_i \epsilon_{l_i}} \quad (1.10)$$

We need take account the degeneracy. As the particle are **indistiguishable**, we can consider all the particle as the same, then we only nned to know how to distribuite N same particles in l levels.

A set of $\{n_l\}$ correspond $\frac{N!}{\prod_i n_i!}$ possible values of $\{l_1, \dots, l_N\}$.

$$Z = \sum_{\{n_i\}, \sum_i n_i = N} e^{-\beta E} = \sum_{\{n_i\}, \sum_i n_i = N} \frac{\prod_i n_i!}{N!} e^{-\beta \sum_i n_i \epsilon_i} \quad (1.11)$$

1.2.2. Boltzmann approximation

The Boltzmann approximation is $n_l! \approx 1$. This is when $\rho \Lambda_T^3 \ll 1$ with $\Lambda_T^3 = h/\sqrt{2\pi m K T}$. This approximation is good when $T \uparrow\uparrow$ or/and $\rho \downarrow\downarrow$ (very diluete cases). Under this condition, bosons and fermions **behave the same way**.

With the Boltzmann approximation, the partition function is

$$Z = \sum_i \frac{1}{N!} e^{-\beta \sum_i \epsilon_{l_i}} = \frac{Z_1^N}{N!} \quad (1.12)$$

1.2.3. Particles in a 3D potential well

Let's calculate the partition function in this system. Let's assume that

$$Z = \sum \frac{1}{N!} (Z_1)^N \quad (1.13)$$

Then let's calculate Z_1

A single particle's energy has two contribution, one due to the traslational degrees of freedom and one due to the internal degrees of freedom.

$$\epsilon = \epsilon_{tr} + \epsilon_{in} \quad (1.14)$$

Then the particle function of a single particle is

$$Z_1 = \sum_{l_{tr}} \sum_{l_{in}} e^{-\beta \epsilon_{tr} - \beta \epsilon_{in}} = Z_1^{tr} Z_1^{in} \quad (1.15)$$

In RAW we developed the solution of Schrödinger's Equation in a potential well in 1D. From this solution we obtained the expression for energy, which is related to the momentum.

$$\epsilon = \frac{nh}{4Lm} \quad (1.16)$$

Using this for the traslational energy expression

$$\epsilon_{tr} = \frac{p_x^2 + p_y^2 + p_z^2}{2m} = \frac{\Lambda}{2m} \left(\frac{n_x^2 h^2}{4L^2} + \frac{n_y^2 h^2}{4L^2} + \frac{n_z^2 h^2}{4L^2} \right) \quad (1.17)$$

Let's calculate the partition function

$$Z_1^{tr} = \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} e^{-\beta \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2)} \quad (1.18)$$

Let's take a look how to solve individually

$$\sum_{n=1}^{\infty} e^{-\beta \frac{h^2}{8mL^2} n^2} \quad (1.19)$$

If we notice, p is related to the length of the well: $p = hn/2L$, then $\Delta p = p_{n+1} - p_n = h/2L$. If we consider a very large L , this is $L \rightarrow \infty$ then $\Delta p = dp$, let's use this to simplify the summatory. Let's change of variable: $n = pL/h\pi$

$$\sum_{n=1}^{\infty} e^{-\beta \frac{h^2}{8mL^2} n^2} = \sum_{p=\frac{h}{2L}}^{\infty} e^{-\beta \frac{p^2}{2m}} \quad (1.20)$$

We introduce Δp in the summatory and we are going to perform a limit, then we can covert the summatory into an integral.

$$\lim_{\rightarrow \infty} \sum_{p=\frac{h}{2L}}^{\infty} \frac{2L}{h} \Delta p e^{-\beta \frac{p^2}{2m}} = \int_0^{\infty} e^{-\beta \frac{p^2}{2m}} dp = \frac{1}{2} \int_{-\infty}^{\infty} e^{-\beta \frac{p^2}{2m}} dp \quad (1.21)$$

Substituting the partition function with this integral we have

$$\frac{1}{h^3} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} dp_x dp_y dp_z e^{-\frac{\beta}{2m}(p_x^2 + p_y^2 + p_z^2)} \quad (1.22)$$

Taking into account the Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}} dx = \sqrt{2\pi\sigma^2} \quad (1.23)$$

It follows that:

$$Z_1^{(tr)} = \frac{L^3}{h^3} (2\pi m k T)^{3/2} = \frac{V}{\Lambda_T^3} \quad (1.24)$$

Then the partition function is

$$Z = \frac{1}{N!} \left(\frac{V}{\Lambda_T^3} \right)^N \quad (1.25)$$

The previous result coincide with the classical ideal monoatomic gas.

1.2.4. Thermodynamic properties of the BG

$$A(T, V, N) = -kT \ln Z_{BG} = -NkT \ln z_1^{(tr)} + \underbrace{kT \ln N!}_{NkT \ln N - NkT} - NkT \ln z_1^{(in)}$$

Using the *Stirling approximation* and knowing that $z_1^{(tr)} = \frac{V}{\Lambda_T^3}$:

$$A(T, V, N) = -NkT + NkT \ln \left(\frac{V}{N} \right) - NkT \ln \Lambda_T^3 + NkT \ln z_1^{(in)}$$

Defining the free energy per particle $a(T, n) = \frac{A(T, V, N)}{N} = -kT + kT \ln(n \Lambda_T^3) + a_{in}(T)$, where $a_{in}(T) = -kT \ln z_1^{(in)}$ and $n = \frac{N}{V}$.

Considering $dA = -SdT - PdV + \mu dN$:

Eq. of State (Pressure)

$$P = - \left(\frac{\partial A}{\partial V} \right)_{T, N} \quad (1.26)$$

Since $-kT \ln z_1^{(in)}$ has no V dependence, only the translational degrees of freedom

contribute to the equation of state (pressure):

$$P = -N \left(-\frac{\partial}{\partial V} (kT \ln(\frac{V}{N})) \right) = \frac{NkT}{V} \implies \text{Boltzmann gas behaves as an ideal gas} \quad (1.27)$$

The internal partition function $z^{(in)}$ contributes to the entropy, free energy, and specific heat.

Entropy

$$S = - \left(\frac{\partial A}{\partial T} \right)_{V,N} \implies s = - \left(\frac{\partial a}{\partial T} \right)_n = - \left[k \ln(n\Lambda_T^3) - k + a'_{in}(T) + \frac{kT \cdot 3\Lambda_T^2 \cdot \Lambda'_T(T)}{\Lambda_T^3} \right] \quad (1.28)$$

Given that $\Lambda_T = \frac{h}{\sqrt{2\pi mkT}} \implies \Lambda'_T(T) = -\frac{\Lambda_T(T)}{2T}$, we derive the **Sackur-Tetrode equation**:

$$s(T, n) = k \left[\frac{5}{2} - \ln(n\Lambda_T^3) \right] - a'_{in}(T) \quad (1.29)$$

It appears that $s(T \rightarrow 0, n) \neq 0$. This occurs because we are using the BG model; therefore, T must be high and/or n must be low for the approximation to hold.

Internal Energy

$$A = U - TS \implies U = A + TS \implies u = a + Ts \quad (1.30)$$

$$u = a + Ts = \frac{3kT}{2} + a_{in}(T) - Ta'_{in}(T) \quad (1.31)$$

The $\frac{3kT}{2}$ term is the contribution from the equipartition theorem. We have a contribution of $\frac{1}{2}kT$ for each quadratic term in the Hamiltonian $H = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} + \dots$

Specific Heat

$$c_v = \left(\frac{\partial u}{\partial T} \right)_n = \frac{3}{2}k - Ta''_{in}(T) \quad (1.32)$$

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1.2.5. Role played by internal degrees of freedom

$$A_{in} = -NkT \ln z_1^{(in)} \quad (\text{Free energy})$$

$$S_{in} = - \left(\frac{\partial A_{in}}{\partial T} \right)_{V,N} = Nk \left(\ln z_1^{(in)} + T \frac{\partial}{\partial T} \ln z_1^{(in)} \right) \quad (\text{Entropy})$$

$$U_{in} = A_{in} + TS_{in} = -NkT \ln z_1^{(in)} + NkT \left(\ln z_1^{(in)} + T \frac{\partial}{\partial T} \ln z_1^{(in)} \right) = NkT^2 \frac{\partial}{\partial T} \ln z_1^{(in)} \quad (\text{Internal energy})$$

$$C_v^{in} = \frac{\partial}{\partial T} \left[NkT^2 \frac{\partial}{\partial T} \ln z_1^{(in)} \right]_{V,N} \quad (\text{Specific heat})$$

If $z_1^{(in)} = z_{rot} \cdot z_{vib} \cdot z_{elec}$:

$$A^{(in)} = -kT \ln z^{(in)} = A_{rot} + A_{vib} + A_{elec}$$

$$S^{(in)} = - \left(\frac{\partial A^{(in)}}{\partial T} \right) = S_{rot} + S_{vib} + S_{elec}$$

$$U^{(in)} = U_{rot} + U_{vib} + U_{elec}$$

$$C_v^{(in)} = C_v^{rot} + C_v^{vib} + C_v^{elec}$$

Activation of internal energy d.o.f.

$$z_1^{(in)} = \sum_n e^{-\beta \varepsilon_n^{(in)}} \quad \text{where } n \text{ is the quantum number.}$$

Spectrum of the internal d.o.f. for the internal d.o.f.

- $\Delta \varepsilon > \varepsilon_1 - \varepsilon_0$ (gap or activation energy)
- $T_a = \frac{\Delta \varepsilon}{k}$ (activation temperature)

First limit case: Thermal energy (kT) much smaller than $\Delta \varepsilon$

$kT \ll \Delta \varepsilon \implies \beta \Delta \varepsilon \gg 1$. Taking $\varepsilon_0, \varepsilon_1, \varepsilon_2 \dots$ and assuming $\ln e^{-\beta \varepsilon_0} = -\beta \varepsilon_0 = -\frac{\varepsilon_0}{kT}$:

$$z^{in} = \sum_{\varepsilon} g(\varepsilon) e^{-\beta \varepsilon} \approx g(\varepsilon_0) e^{-\beta \varepsilon_0} \left(1 + \frac{g(\varepsilon_1)}{g(\varepsilon_0)} e^{-\beta \Delta \varepsilon} + \dots \right) \approx e^{-\beta \varepsilon_0} g(\varepsilon_0) \quad (1.33)$$

Then $C_v^{in} = k \frac{\partial}{\partial T} \left[T^2 \frac{\partial}{\partial T} \ln z^{in} \right] \approx k \frac{\partial}{\partial T} \left[T^2 \frac{\partial}{\partial T} \left(-\frac{\varepsilon_0}{kT} \right) \right] = k \frac{\partial}{\partial T} \left[T^2 \frac{\varepsilon_0}{kT^2} \right] = 0. \implies$ In this case, internal degrees of freedom do not contribute to C_v .

Second limit case: Thermal energy (kT) much larger than $\Delta \varepsilon$

$kT \gg \Delta \varepsilon$: We can take the continuous limit of the specific energies and the system behaves as classical. $\implies$ Equipartition theorem holds $\implies U = \alpha NkT$ ($C_v = \text{constant}$).

Graph of $C_v = C_v^{(tr)} + C_v^{(in)}$

- At low T : $C_v^{in} = 0$, so $C_v = \frac{3}{2}k$.
- At high T ($kT \gg \Delta \varepsilon$): C_v increases and reaches a new plateau due to the activation of internal degrees of freedom.